MULTIVARIABLE CALCULUS • UNIT 6.01

C Vector Fields, Conservative Fields & Line Integrals

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 D	Q4 D	Q5 A
Q6 D	Q7 A	Q8 C	Q9 C	Q10 A
Q11 A	Q12 C	Q13 A	Q14 B	Q15 B
Q16 C	Q17 B	Q18 C	Q19 A	Q20 D
Q21 A	Q22 B	Q23 B	Q24 D	Q25 D

Vector Fields, Conservative Fields & Line Integrals Practice Exercises

1 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

- **A** $\langle 6, 8 \rangle$ ✓
- **B** $\langle 8, 6 \rangle$
- **C** $\langle 5, 6 \rangle$
- **D** $\langle 6, 5 \rangle$

WORKED SOLUTION & VERIFICATION

The vector field is given by $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. To evaluate the vector field at the point $(2, 4)$, we substitute $x = 2$ and $y = 4$ into the components of the vector field.
First component: $3x = 3(2) = 6$.
Second component: $2y = 2(4) = 8$.
Therefore, $\mathbf{F}(2, 4) = \langle 6, 8 \rangle$.

2 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, 3y \rangle$. Which of the following options best represents the vector field at the point (1, 1)?

- **A** $\langle 2, 3 \rangle$ ✓
- **B** $\langle 1, 1 \rangle$
- **C** $\langle 4, 6 \rangle$
- **D** $\langle 2, 6 \rangle$

WORKED SOLUTION & VERIFICATION

To find the representation of the vector field $\mathbf{F}(x, y) = \langle 2x, 3y \rangle$ at a specific point, we substitute the coordinates of the point into the components of the vector field. At the point (1, 1), we have: $x = 1$ and $y = 1$. Substituting these values into $\mathbf{F}(x, y)$ gives: $\mathbf{F}(1, 1) = \langle 2(1), 3(1) \rangle = \langle 2, 3 \rangle$. Therefore, the vector field at the point (1, 1) is $\langle 2, 3 \rangle$. Option B is incorrect as it represents the position vector at (1,1). Option C is incorrect as it represents the vector field at (2,2). Option D is incorrect as it uses the x-component from (2,y) and the y-component from (x,2).

3 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3, -1)$?

A $\langle 5, 6 \rangle$

B $\langle 5, 0 \rangle$

C $\langle 7, 6 \rangle$

D $\langle 7, 0 \rangle$ ✓

WORKED SOLUTION & VERIFICATION

To find the value of the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$ at the point $(3, -1)$, we substitute $x = 3$ and $y = -1$ into the components of the vector field.

The first component is $2x + y = 2(3) + (-1) = 6 - 1 = 5$.

The second component is $x - 3y = 3 - 3(-1) = 3 + 3 = 6$.

Therefore, the vector field at $(3, -1)$ is $\langle 5, 6 \rangle$.

4 MCQ

Evaluate $\int_C x^2 \, ds$ where C is the segment from $(0,\ 0)$ to $(5,\ 12)$.

A $\frac{25}{3}$

B $\frac{4225}{3}$

C $\frac{328}{3}$

D $\frac{325}{3}$ ✓

WORKED SOLUTION & VERIFICATION

$\mathbf{r}(t) = \langle 5t, 12t \rangle,\ ds = 13\, dt.\ \int_0^1 325t^2 \, dt = \frac{325}{3}.$

5 MCQ

Which vector field is conservative on all of R^2?

A $\langle e^y, xe^y \rangle$ ✓

B $\langle xy, x \rangle$

C $\langle x + y, -x + y \rangle$

D $\langle x^2 y, xy^2 \rangle$

WORKED SOLUTION & VERIFICATION

On R^2, $\langle P, Q \rangle$ is conservative iff $P_y = Q_x$. For $\langle e^y, xe^y \rangle$: $P_y = e^y = Q_x$; each other field has $P_y \not\equiv Q_x$.

6 MCQ

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, 2x \rangle$ along the segment from $(-2,\ -2)$ to $(-2,\ 3)$.

A 20

B -19

C -40

D -20 ✓

WORKED SOLUTION & VERIFICATION

Parametrize $\mathbf{r}(t) = \langle -2, 5t - 2 \rangle,\ 0 \le t \le 1$. Then $\int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_0^1 (-20)\ dt = -20.$

7 MCQ

Evaluate $\int_C x + y\,ds$ where C is the segment from $(0,\ 0)$ to $(4,\ 3)$.

- **A** $\frac{35}{2}$ ✓
- **B** $\frac{7}{2}$
- **C** $\frac{175}{2}$
- **D** $\frac{37}{2}$

WORKED SOLUTION & VERIFICATION

$\mathbf{r}(t) = \langle 4t, 3t\rangle,\ ds = 5\,dt.\ \int_0^1 35t\,dt = \frac{35}{2}.$

8 MCQ

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle xy, y\rangle$ along the segment from $(1,\ 1)$ to $(1,\ -2)$.

- **A** $-\frac{3}{2}$
- **B** $\frac{5}{2}$
- **C** $\frac{3}{2}$ ✓
- **D** 3

WORKED SOLUTION & VERIFICATION

Parametrize $\mathbf{r}(t) = \langle 1, 1 - 3t\rangle,\ 0 \le t \le 1$. Then $\int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt = \int_0^1 (9t - 3)\ dt = \frac{3}{2}$.

9 MCQ

$\mathbf{F} = \langle ye^x, e^x\rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(2,\ 0)$ to $(1,\ -1)$.

- **A** $1 - e$
- **B** $-2e$
- **C** $-e$ ✓
- **D** e

WORKED SOLUTION & VERIFICATION

A potential is $f = ye^x$. By the Fundamental Theorem of Line Integrals the value is $f(1,\ -1) - f(2,\ 0) = -e$.

10 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy\rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ 0)$ to $(-2,\ 3)$.

- **A** -20 ✓
- **B** 20
- **C** -19
- **D** -40

WORKED SOLUTION & VERIFICATION

A potential is $f = xy^2 + x$. By the Fundamental Theorem of Line Integrals the value is $f(-2,\ 3) - f(0,\ 0) = -20$.

11 MCQ

$\mathbf{F} = \langle y, x+z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 2,\ 0)$ to $(-1,\ -1,\ 3)$.

- **A** -4 ✓
- **B** 4
- **C** -2
- **D** -3

WORKED SOLUTION & VERIFICATION

A potential is $f = xy + yz$. By the Fundamental Theorem of Line Integrals the value is $f(-1,\ -1,\ 3) - f(1,\ 2,\ 0) = -4$.

12 MCQ

$\mathbf{F} = \langle 2x+y, x+3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-1,\ 0)$ to $(0,\ -2)$.

- **A** -8
- **B** -18
- **C** -9 ✓
- **D** 9

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2 + xy + y^3$. By the Fundamental Theorem of Line Integrals the value is $f(0,\ -2) - f(-1,\ 0) = -9$.

13 MCQ

$\mathbf{F} = \langle 2xy, x^2 - 2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ 0)$ to $(1,\ 3)$.

- **A** -3 ✓
- **B** 3
- **C** -2
- **D** -6

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2y - 2y$. By the Fundamental Theorem of Line Integrals the value is $f(1,\ 3) - f(-2,\ 0) = -3$.

14 MCQ

$\mathbf{F} = \langle 2x+y, x+3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ 0)$ to $(0,\ 0)$.

- **A** 4
- **B** -4 ✓
- **C** 0
- **D** -3

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2 + xy + y^3$. By the Fundamental Theorem of Line Integrals the value is $f(0,\ 0) - f(-2,\ 0) = -4$.

15 MCQ

$\mathbf{F} = \langle 2x + 3y, 3x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, -1)$ to $(2, -2)$.

A -16

B -13 ✓

C -12

D 13

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2 + 3xy + y^3$. By the Fundamental Theorem of Line Integrals the value is $f(2, -2) - f(1, -1) = -13$.

16 MCQ

$\mathbf{F} = \langle y, x + z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, 0, 0)$ to $(-2, 2, 3)$.

A -2

B 3

C 2 ✓

D 4

WORKED SOLUTION & VERIFICATION

A potential is $f = xy + yz$. By the Fundamental Theorem of Line Integrals the value is $f(-2, 2, 3) - f(1, 0, 0) = 2$.

17 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, 0)$ to $(2, 3)$.

A -19

B 19 ✓

C 20

D 38

WORKED SOLUTION & VERIFICATION

A potential is $f = xy^2 + x$. By the Fundamental Theorem of Line Integrals the value is $f(2, 3) - f(1, 0) = 19$.

18 MCQ

$\mathbf{F} = \langle 2x + 2y, 2x + 3y^2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1, -1)$ to $(-2, 1)$.

A -3

B 1

C 3 ✓

D 4

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2 + 2xy + y^3$. By the Fundamental Theorem of Line Integrals the value is $f(-2, 1) - f(1, -1) = 3$.

19 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x)\rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ 1)$ to $(0,\ -2)$.

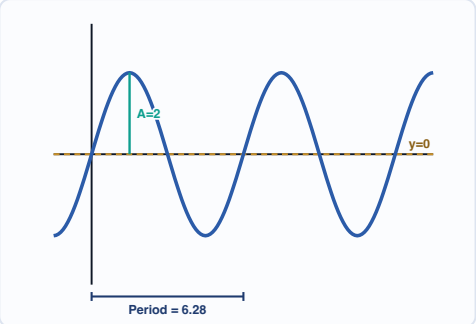
- **A** $\sin(2) + 2$ ✓
- **B** $-2 - \sin(2)$
- **C** 0
- **D** $\sin(2) + 3$

WORKED SOLUTION & VERIFICATION

A potential is $f = x + y\sin(x)$. By the Fundamental Theorem of Line Integrals the value is $f(0,\ -2) - f(-2,\ 1) = \sin(2) + 2$.

20 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x)\rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(-2,\ -1)$ to $(-2,\ 1)$.



- **A** $2\sin(2)$
- **B** $-2 - \sin(2)$
- **C** $1 - 2\sin(2)$
- **D** $-2\sin(2)$ ✓

WORKED SOLUTION & VERIFICATION

A potential is $f = x + y\sin(x)$. By the Fundamental Theorem of Line Integrals the value is $f(-2,\ 1) - f(-2,\ -1) = -2\sin(2)$.

21 MCQ

$\mathbf{F} = \langle 2x + 3y, 3x + 3y^2\rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 2)$ to $(-1,\ 0)$.

- **A** -14 ✓
- **B** 14
- **C** 1
- **D** -13

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2 + 3xy + y^3$. By the Fundamental Theorem of Line Integrals the value is $f(-1,\ 0) - f(1,\ 2) = -14$.

22 MCQ

$\mathbf{F} = \langle y^2 + 1, 2xy \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ -1)$ to $(3,\ 2)$.

A 14	B 13 ✓
C -13	D 15

WORKED SOLUTION & VERIFICATION

A potential is $f = xy^2 + x$. By the Fundamental Theorem of Line Integrals the value is $f(3,\ 2) - f(1,\ -1) = 13$.

23 MCQ

$\mathbf{F} = \langle 2xy, x^2 - 2 \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(1,\ 0)$ to $(2,\ -2)$.

A 4	B -4 ✓
C -3	D -8

WORKED SOLUTION & VERIFICATION

A potential is $f = x^2y - 2y$. By the Fundamental Theorem of Line Integrals the value is $f(2,\ -2) - f(1,\ 0) = -4$.

24 MCQ

$\mathbf{F} = \langle y, x + z, y \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ -2,\ 1)$ to $(-2,\ 0,\ 2)$.

A -2	B 0
C 3	D 2 ✓

WORKED SOLUTION & VERIFICATION

A potential is $f = xy + yz$. By the Fundamental Theorem of Line Integrals the value is $f(-2,\ 0,\ 2) - f(0,\ -2,\ 1) = 2$.

25 MCQ

$\mathbf{F} = \langle y\cos(x) + 1, \sin(x) \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0,\ 2)$ to $(-2,\ 3)$.

A $2 + 3\sin(2)$	B $-3\sin(2) - 1$
C $-6\sin(2) - 4$	D $-3\sin(2) - 2$ ✓

WORKED SOLUTION & VERIFICATION

A potential is $f = x + y\sin(x)$. By the Fundamental Theorem of Line Integrals the value is $f(-2,\ 3) - f(0,\ 2) = -3\sin(2) - 2$.